Airline Passenger Satisfaction Analysis Using Python

PROJECT OVERVIEW

This project focuses on analyzing airline passenger data using Python.

The main activities performed in this project are:

1. Analyzing airline passenger satisfaction levels.
2. Understanding passenger travel preferences and customer types.
3. Examining service ratings to understand passenger experience.
4. Analyzing departure and arrival delay patterns.
5. Performing data cleaning and exploratory data analysis (EDA).
6. Conducting statistical analysis and data visualization.
7. Identifying key insights to understand factors influencing passenger satisfaction.

The main aim is to provide meaningful insights that can help airlines understand passenger expectations and improve service quality.

DATA SOURCE

The dataset used for this project contains airline passenger information and satisfaction details.

- **Dataset Name:** Airline Passenger Satisfaction Dataset
- **Source:** airline_passenger_satisfaction_Analysis using Python.csv
- **File Type:** CSV
- **Number of Records:** 1,29,880 customer records.
- **Data Format:** Structured Tabular Data

The dataset includes passenger details, travel information, service ratings, flight delays, and satisfaction levels.

PROBLEM STATEMENT

Airlines collect large amounts of passenger data, but understanding passenger satisfaction and identifying the factors that influence passenger experience can be challenging

The key challenges addressed in this project are:

1. Understanding passenger satisfaction levels.
2. Identifying differences in satisfaction based on travel type and passenger class.
3. Analyzing passenger service ratings to understand service quality.
4. Examining departure and arrival delays.
5. Identifying patterns and relationships between passenger details and satisfaction.

This project aims to analyze airline passenger data using Python and generate meaningful insights that can help airlines understand customer expectations and improve passenger experience.

PROJECT OBJECTIVES

The main objectives of this project are:

1. To analyze overall airline passenger satisfaction levels.
2. To identify factors influencing passenger satisfaction.
3. To compare satisfaction levels based on travel type and passenger class.
4. To evaluate passenger ratings for different airline services.
5. To analyze departure and arrival delay patterns.
6. To identify trends and relationships in passenger data.
7. To provide meaningful insights and recommendations to improve airline service quality.

ATTRIBUTE DETAILS

Attribute Name	Description
Gender	Gender of the passenger.
Customer Type	Indicates whether the passenger is a loyal or disloyal customer.
Age	Age of the passenger.
Type of Travel	Business travel or personal travel.
Class	Business, Eco, or Eco Plus class.
Flight Distance	Distance travelled by the passenger.
Inflight Wi-Fi Service	Rating of inflight Wi-Fi service.
Food and Drink	Rating of food and beverage services.
Seat Comfort	Rating of seat comfort.
Inflight Entertainment	Rating of entertainment services.
Cleanliness	Rating of cabin cleanliness.
Online Boarding	Rating of online boarding service.
Departure Delay	Departure delay time in minutes.
Arrival Delay	Arrival delay time in minutes.
Satisfaction	Overall passenger satisfaction level.

TOOLS & TECHNOLOGIES

The following tools and technologies were used in this project:

1. **Python:** Used as the main programming language for data analysis.
2. **Google Colab:** Used to write and execute Python code.
3. **Pandas:** Used for data loading, cleaning, manipulation, and analysis.
4. **NumPy:** Used for numerical operations and calculations.
5. **Matplotlib:** Used to create data visualizations.
6. **Seaborn:** Used to create statistical charts and visualizations.
7. **SciPy:** Used for statistical analysis and hypothesis testing.
8. **GitHub:** Used as the source for accessing the dataset.
9.

DATA PRE-PROCESSING AND FEATURE ENGINEERING

The following steps were considered during data pre-processing and feature engineering:
A. Data Pre-Processing

1. Loaded the airline passenger dataset into Google Colab using Pandas.
2. Inspected the dataset structure, column names, and data types.
3. Checked for missing values in the dataset.
4. Checked for duplicate records.
5. Prepared the cleaned dataset for further analysis using the `df_clean` dataframe.

B. Feature Engineering

1. Created an age group feature to categorize passengers based on their age.

2 .Used the transformed dataset for further exploratory data analysis and visualization.

Age Group Categorization:

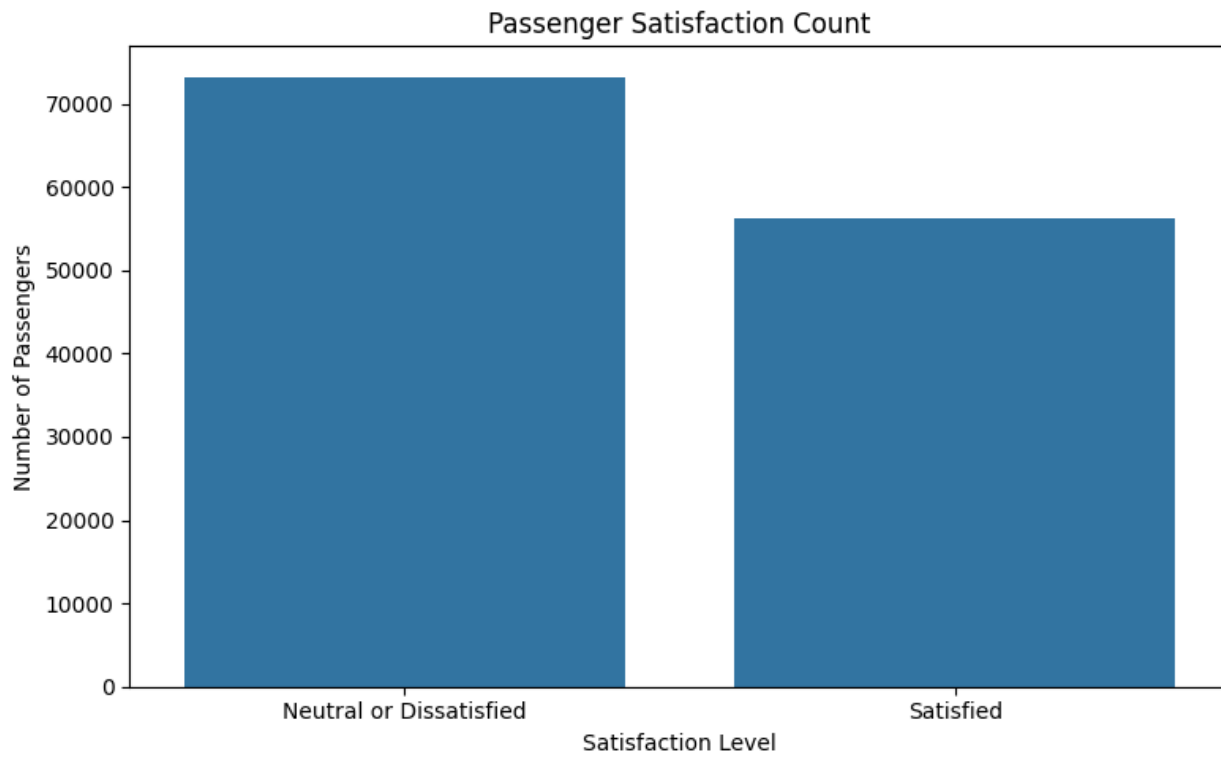
1. Created a new feature named `age_group` based on passenger age.
2. Categorized passengers into different age groups to understand satisfaction patterns across age categories.
3. This helped in comparing passenger satisfaction based on age

ANALYSIS AND VISUALIZATIONS

The airline passenger dataset was analyzed using Python to understand passenger satisfaction and identify important patterns.

The following analyses and visualizations were performed:

Satisfaction Count Plot



Interpretation

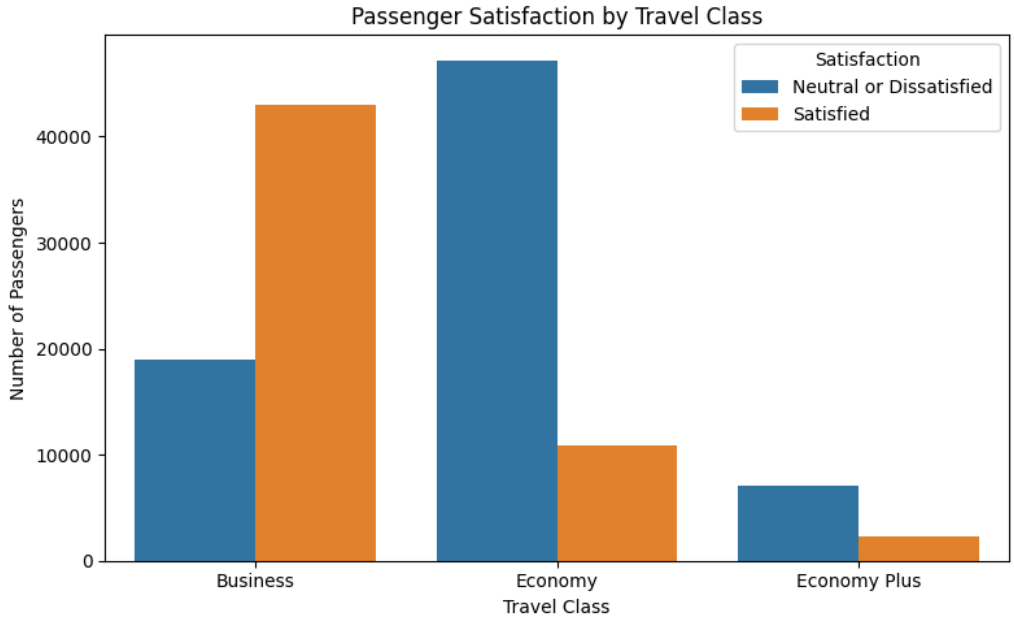
1. The number of neutral or dissatisfied passengers is higher than the number of satisfied passengers.

2. Approximately 73,000 passengers are neutral or dissatisfied, while around 56,000 passengers are satisfied.

3. The results indicate that passenger satisfaction needs improvement.

4. Airlines should focus on improving service quality and overall passenger experience.

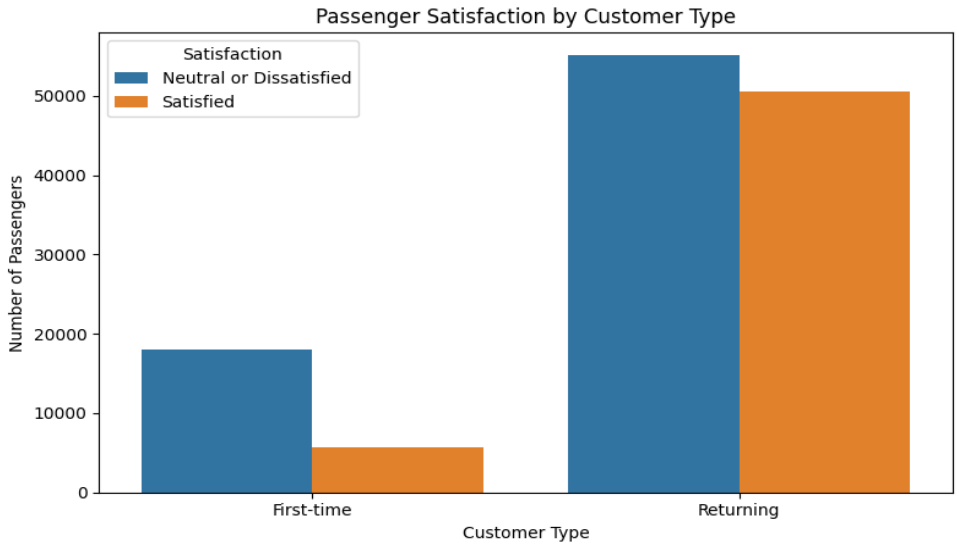
Satisfaction by Travel Class



Interpretation

1. **Business Class** has a higher number of satisfied passengers compared to neutral or dissatisfied passengers.
2. **Economy Class** has a significantly higher number of neutral or dissatisfied passengers.
3. **Economy Plus** also shows more neutral or dissatisfied passengers than satisfied passengers.
4. Business Class passengers show a better satisfaction level compared to other travel classes.

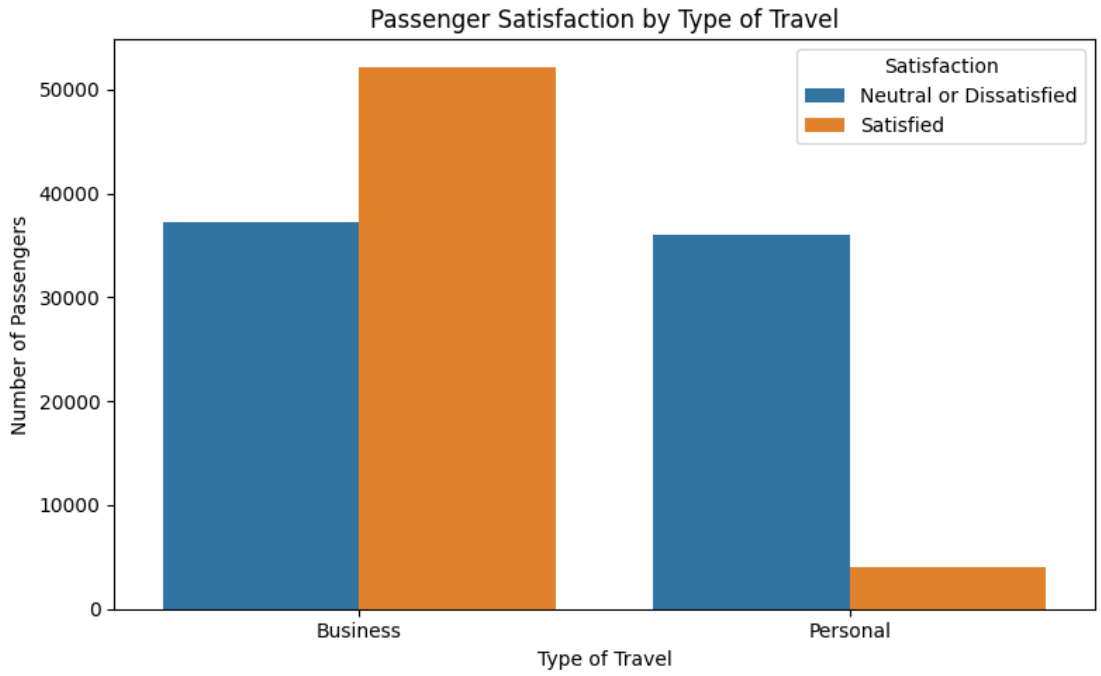
Satisfaction by Customer Type



Interpretation

1. First-time customers have approximately **18,000** neutral or dissatisfied passengers, compared to only **5,700 satisfied passengers**.
2. Returning customers have approximately **55,000** neutral or dissatisfied passengers and **50,500** satisfied passengers.
3. The number of returning customers is significantly higher than first-time customers.
4. Returning customers show a better satisfaction ratio compared to first-time customers.
5. The company should focus on improving the experience of first-time passengers to increase customer satisfaction and customer retention.

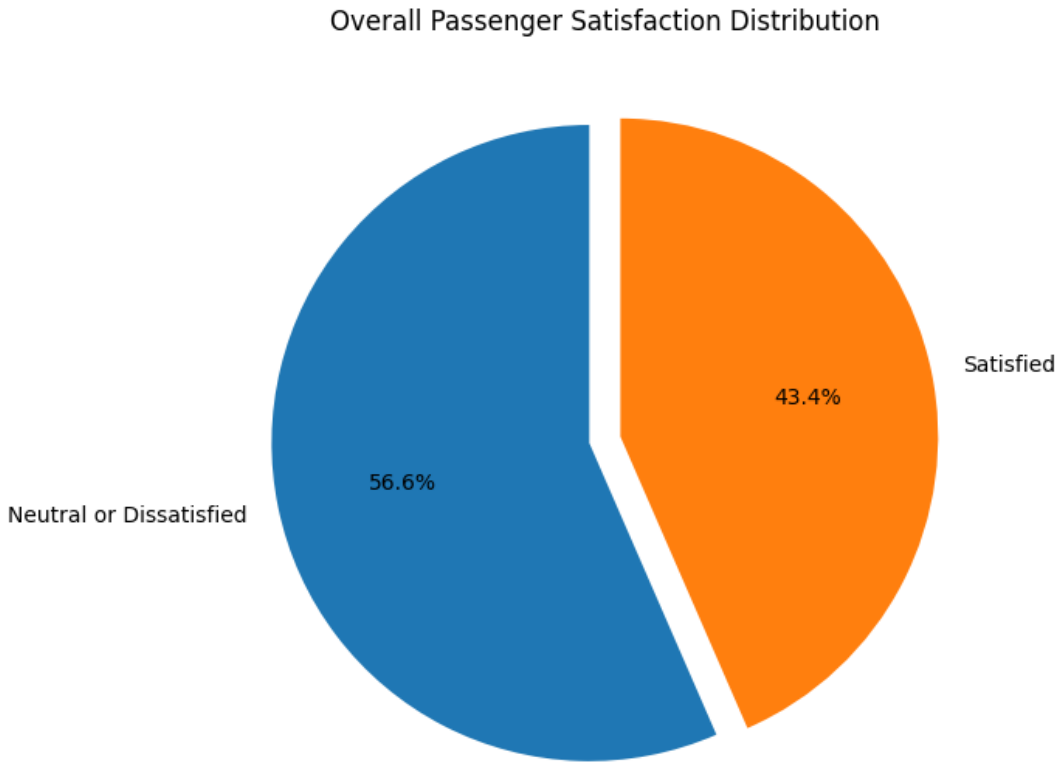
Satisfaction by Type of Travel



Interpretation

1. Business travellers have approximately **52,000 satisfied** passengers, which is higher than neutral or dissatisfied passengers (37,000).
2. Personal travellers have approximately 36,000 neutral or dissatisfied passengers, but only **4,000** satisfied passengers.
3. Business travellers show a higher satisfaction level compared to personal travellers.
4. Personal travellers have a significantly higher dissatisfaction level, indicating poor travel experience.
5. The airline should focus on improving services for personal travellers to increase customer satisfaction and improve overall passenger experience.

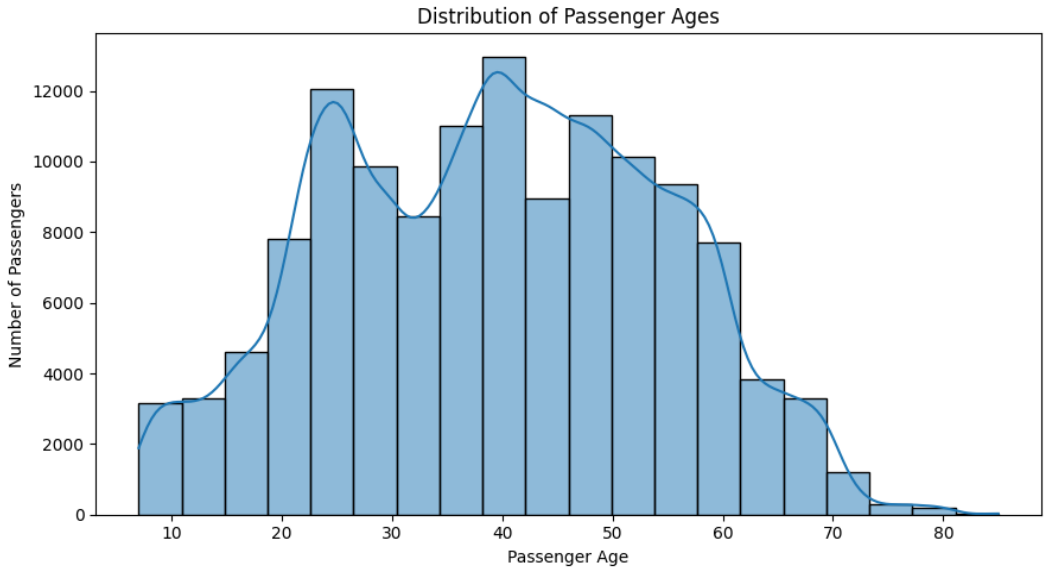
Overall Satisfaction Percentage



Interpretation

- **56.6%**of passengers are neutral or dissatisfied, indicating that more than half of the passengers are not fully satisfied with the airline services.
- **43.4%** of passengers are satisfied with the airline services.
- Overall, passenger dissatisfaction is higher than satisfaction.
- The airline should focus on improving **Service Quality** and passenger experience to increase customer satisfaction.

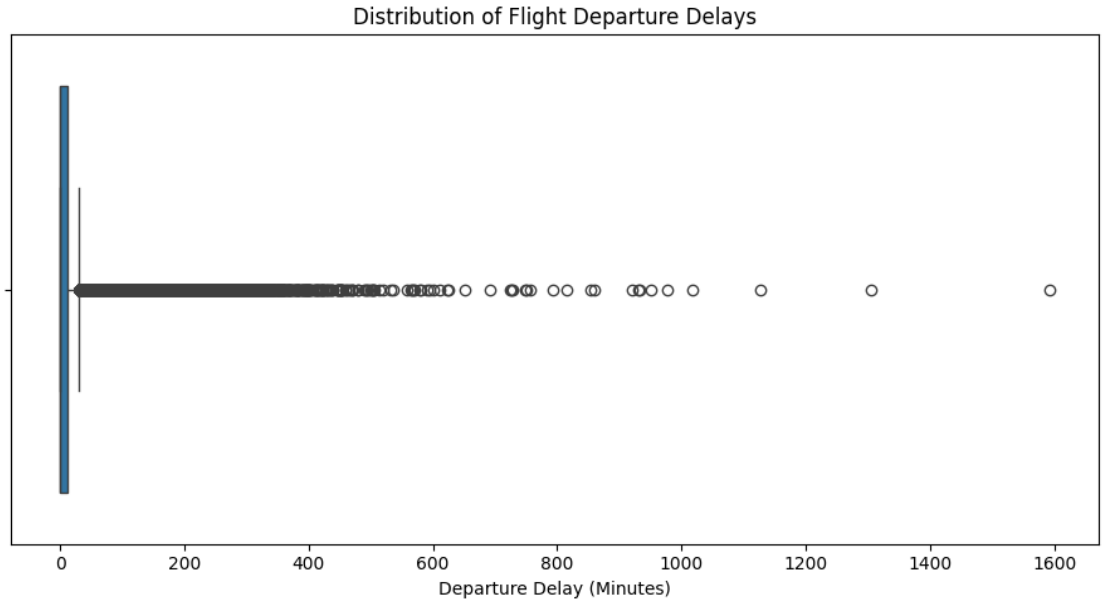
Passenger Age Distribution



Interpretation

- The majority of passengers are between **20 and 60 years** old, indicating that middle-aged people make up a large portion of travellers.
- The highest number of passengers belongs to the **35–45 age** group, with the peak around 40 years.
- The number of passengers gradually **decreases after 60 years**, indicating that fewer senior citizens travel compared to younger and middle-aged passengers.

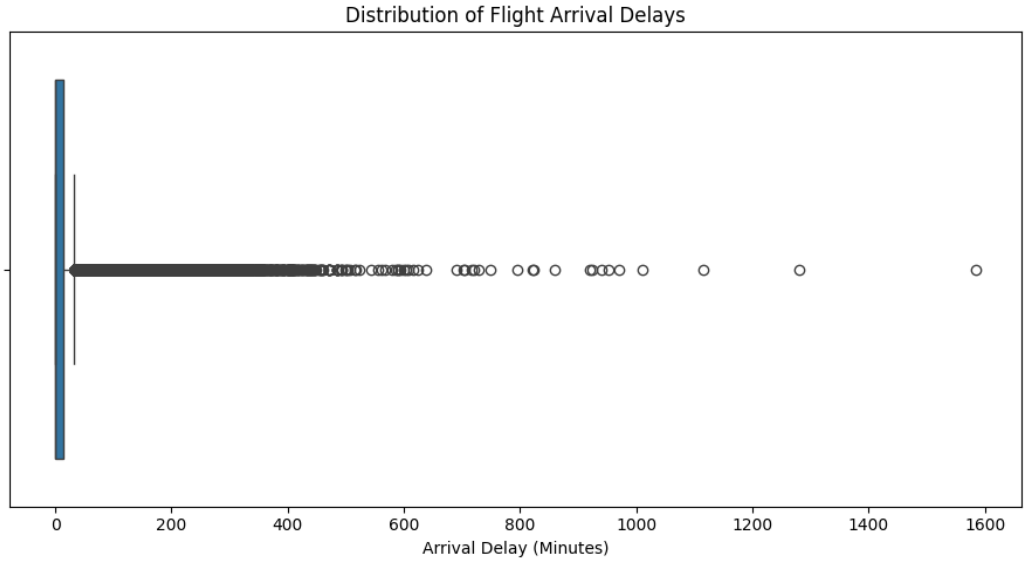
Departure Delay Distribution



Interpretation

1. Most flights have very low departure delays, indicating that the majority of flights depart on time or with minimal delays.
2. The box plot shows several extreme outliers, with some flights experiencing delays of more than 1,000 minutes.
3. The distribution is highly right-skewed, indicating that a small number of flights experience very long delays. The airline should focus on reducing these delays to improve passenger satisfaction.

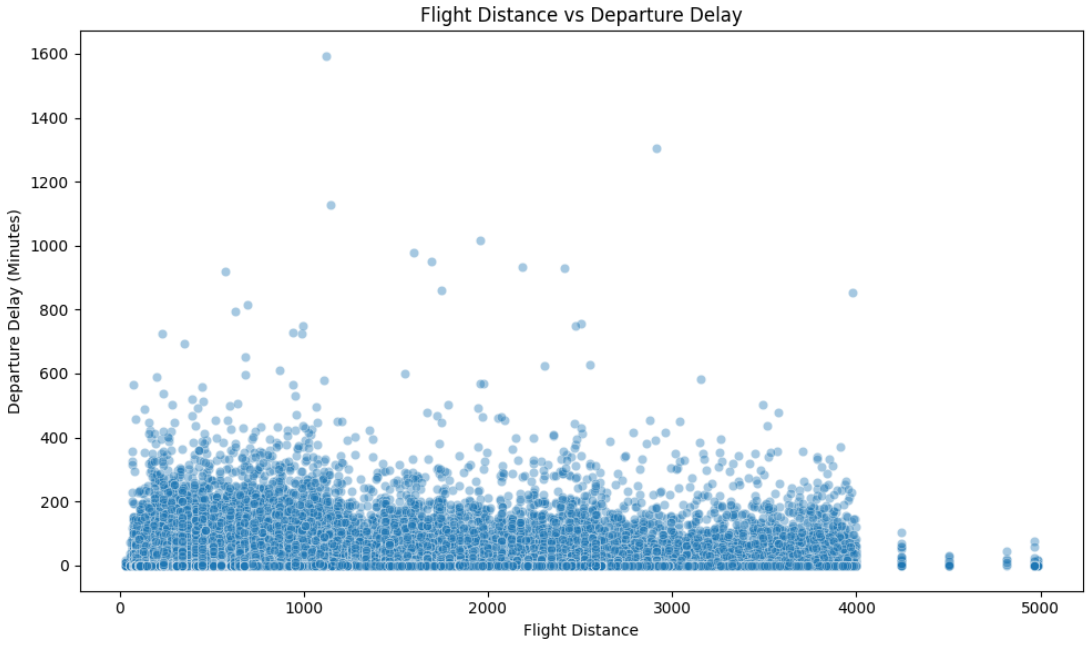
Arrival Delay Distribution



Interpretation

1. The distribution is positively skewed (right-skewed), indicating that most flights have low arrival delays.
2. Most arrival delays are concentrated near zero, while a few flights experience very long delays.
3. Several extreme outliers are present, with the maximum delay reaching approximately 1,600 minutes, indicating occasional severe flight delays.

Flight Distance vs Departure Delay



Interpretation

1. The scatter plot shows no strong correlation between flight distance and departure delay.
2. Most flights experience departure delays of less than 200 minutes, regardless of flight distance.
3. A few flights show extremely high delays, reaching approximately 1,600 minutes, indicating the presence of outliers.
4. Flight delays are widely distributed across different distances, suggesting that flight distance alone does not determine departure delays.s.

Key Findings

Based on the Exploratory Data Analysis and visualizations, the following key findings were identified:

1. **Overall Satisfaction:** 56.6% of passengers were neutral or dissatisfied, while 43.4% were satisfied. This indicates scope for improving the overall passenger experience.
2. **Customer Type:** Returning customers showed a higher satisfaction count compared to first-time customers. First-time passengers may need more attention to improve their travel experience.

3. **Travel Type:** Business travellers showed a higher satisfaction count, whereas personal travellers had more neutral or dissatisfied responses.
4. **Travel Class:** Business class passengers demonstrated a better satisfaction pattern compared to Economy and Economy Plus passengers.
5. **Age Distribution:** Most passengers belonged to the 20–60 age group, with passenger numbers highest around the age of 40.
6. **Flight Delays:** Both arrival and departure delays showed right-skewed distributions, with a few extreme delay values.
7. **Flight Distance:** No strong visible relationship was observed between flight distance and departure delay.
8. **Service Quality:** Passenger ratings across different airline services helped identify areas for comparing service performance and understanding passenger experience.

Overall Finding

The analysis indicates that passenger satisfaction varies across customer types, travel categories, and travel classes. Improving services for personal travellers, first-time passengers, and economy-class passengers may help enhance overall passenger satisfaction.

Business Recommendations

Based on the analysis and key findings, the following business recommendations are suggested to improve airline passenger satisfaction.

1. Improve Economy Class Services

- Focus on improving the overall passenger experience in Economy and Economy Plus classes.
- Enhance seating comfort, onboard services, and passenger convenience.

2. Enhance First-Time Passenger Experience

- Provide better guidance and assistance to first-time passengers.
- Improve check-in support, boarding procedures, and travel information.

3. Improve Personal Travel Experience

- Focus on the needs of personal travellers.
- Provide comfortable seating, quality food services, and better onboard facilities.

4. Improve Flight Delay Management

- Monitor departure and arrival delays regularly.
- Improve flight scheduling and provide timely updates to passengers about delays.

5. Enhance Service Quality

- Regularly monitor passenger ratings for Wi-Fi, food, seat comfort, entertainment, cleanliness, and online boarding.
- Identify services with lower ratings and take appropriate improvement measures.

6. Strengthen Customer Retention

- Maintain good service quality for returning customers.
- Introduce customer loyalty benefits and personalized travel services.

7. Monitor Passenger Satisfaction

- Conduct regular passenger feedback surveys.
- Use satisfaction analysis to identify service gaps and support better business decisions.

Future Enhancements

1. Develop an interactive dashboard using Power BI.
2. Collect more passenger feedback to understand customer expectations.
3. Perform detailed analysis of flight delays.
4. Improve airline services based on passenger ratings.
5. Update the dataset regularly with new passenger information.
6. Analyze passenger preferences based on different travel categories.
7. Identify low-rated services and suggest improvements.
8. Generate regular reports to support better business decisions.

Conclusion

The Airline Passenger Satisfaction Analysis project was successfully completed using Python. The project involved data cleaning, preprocessing, feature engineering, exploratory data analysis, statistical analysis, and data visualization.

The analysis helped identify important factors influencing passenger satisfaction, including customer type, travel type, travel class, service ratings, and flight delays.

The findings showed that 56.6% of passengers were neutral or dissatisfied, while 43.4% were satisfied. The analysis also highlighted opportunities to improve Economy Class services, personal travel experience, and flight delay management.

Overall, this project provided useful insights into passenger behaviour and airline service quality. It demonstrates how Python-based data analysis can support better business decisions and help improve passenger satisfaction.